Peng 2019 PDAC — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 16 samples intermix on one continuous connected component (0 detached islands; UMAP aspect 1.17; iLISI median 1.0→5.3 of 16 at θ=10) — judged on the embedding, not a scalar. Cell cycle (6.6% cycling) is not a module confound (|ρ|≤0.24; the pSom–Crest spatial anti-correlation is unchanged when cyclers are excluded), so the whole malignant manifold is carried forward.

Gate 2 — module geography → NO-GO

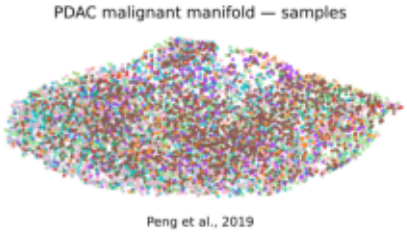
Embryonic module pattern absent. The three smoothed modules occupy a 2-D triangle (PCA variance 0.55/0.40; Gast/pSom/Crest top-decile peaks 110–134° apart) — no Gast/pSom co-expression hub. They do not order along pseudotime (pSom ρ=+0.11, Crest ρ=−0.10) and Crest is non-terminal (Crest-high decile at pseudotime 0.60; joint pSom&Crest enrichment 0.41). Neither the branching hub nor the non-branching pSom→Crest axis reproduces.

Gate 3 — hub cross-repression → NA

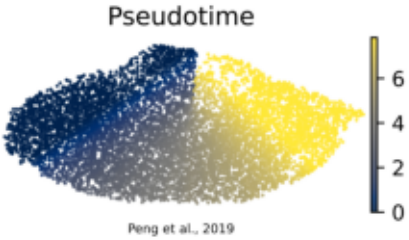
Branching-only gate. The manifold is non-branching (curve: 2 tips, 0 forks) and Gate 2 did not pass. No hub — Gast and pSom are anti-correlated and sit at opposite triangle corners (spatial r=−0.41).

STEP FIGURES

Manifold — samples



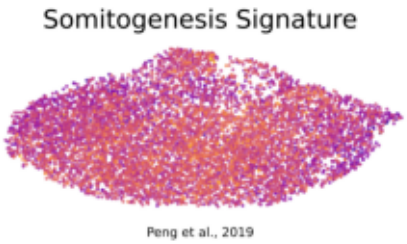
Pseudotime



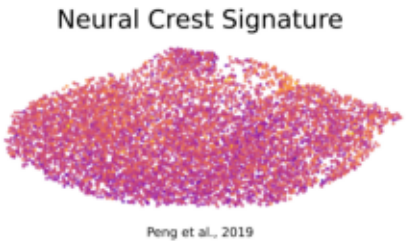
Gastrulation



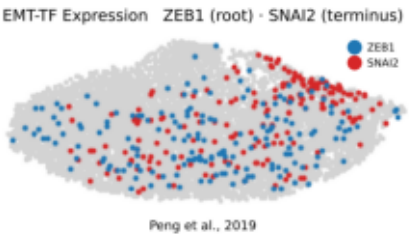
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Peng et al., 2019), 10x — 7,945 malignant cells, 16 samples.

Verdict.
Pattern does NOT reproduce. Gate 1 GO / Gate 2 NO-GO / Gate 3 NA — developmental programs are present as scores but geographically unresolved (a triangle, not an axis): a triangular, non-axial module geography, not the Linear-pSom/Crest developmental axis seen in the same-tissue Steele/Lin/Moncada PDAC GO runs. EMT-TFs ZEB1 (root end) / SNAI2 (terminus end).

Parameter choices

Threshold	≥100-cell floor (gap-finding premise unmet) → all 16 samples kept	**Integration**	Harmony θ=10; iLISI median 1.0→5.3 of 16
Cell cycle	6.6% cycling; not a module confound (|ρ|≤0.24); whole manifold carried	**Module geometry**	2-D triangle (PCA var 0.55/0.40; peaks 110–134° apart); no hub
Trajectory	scFates curve (2 tips, 0 forks); root = Gast/pSom-high tip	**EMT-TFs**	ZEB1 (root) / SNAI2 (terminus); Snail/Zeb/Twist + FOXC/PRRX, 3% floor